

Операционные системы

Командная оболочка Midnight Commander

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

2. Процесс выполнения лабораторной работы

Работа с Midnight Commander

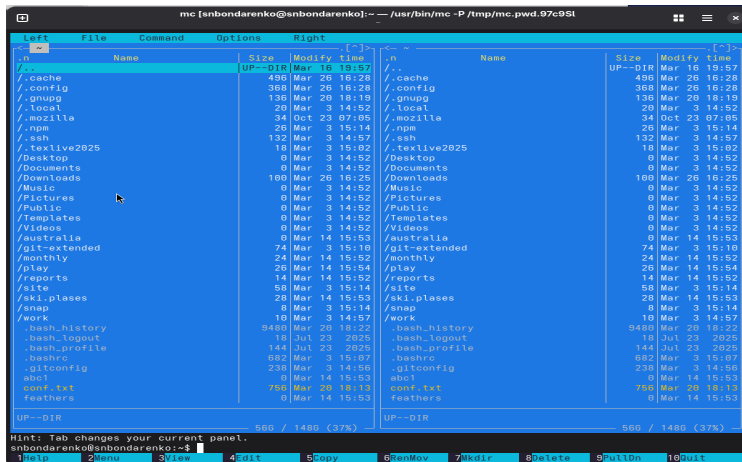


Рисунок 1: Запуск mc

Работа с Midnight Commander

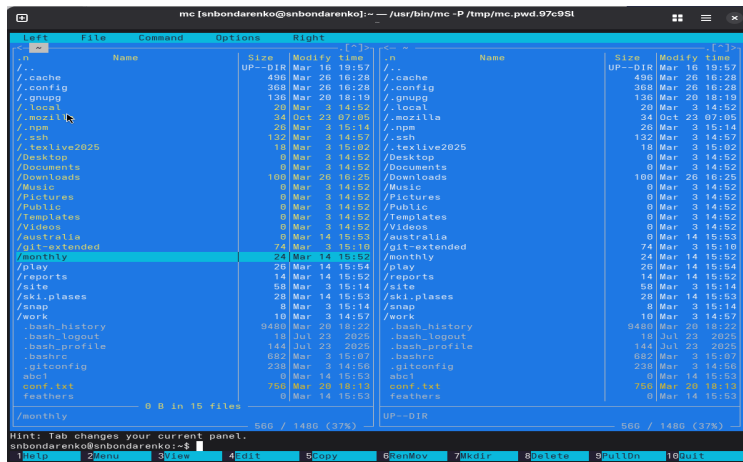


Рисунок 2: Выделение

Работа с Midnight Commander

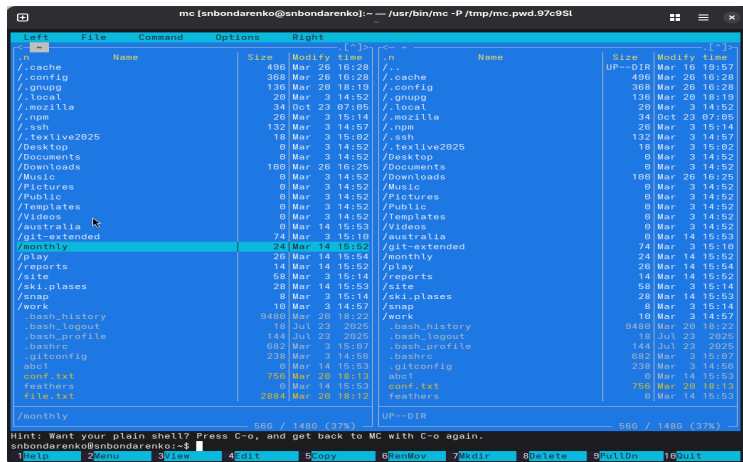


Рисунок 3: Отмена

Работа с Midnight Commander

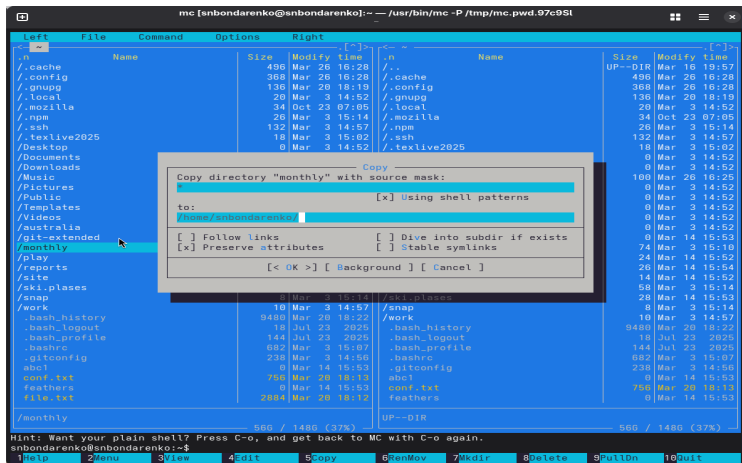


Рисунок 4: Копирование

Работа с Midnight Commander

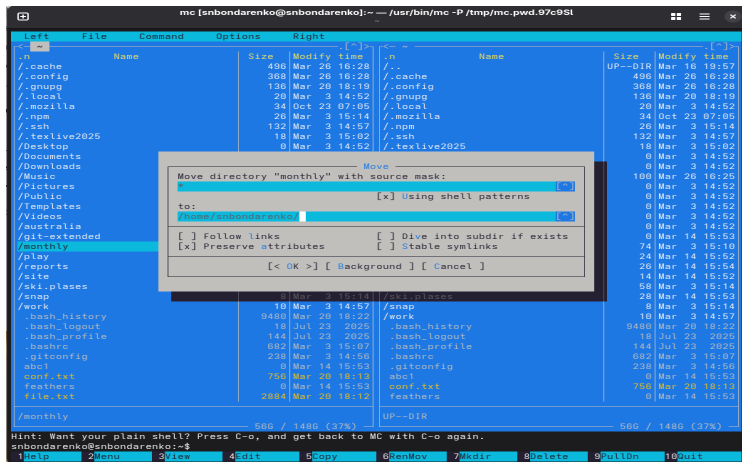


Рисунок 5: Перемещение

Работа с Midnight Commander

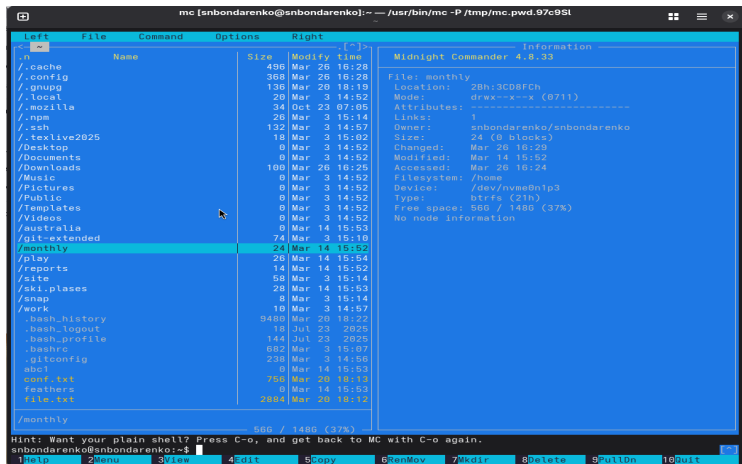


Рисунок 6: Информация

Работа с Midnight Commander

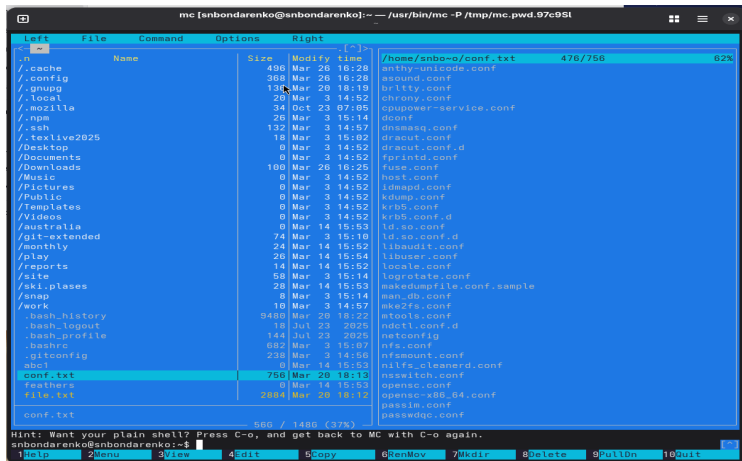


Рисунок 7: Быстрый просмотр

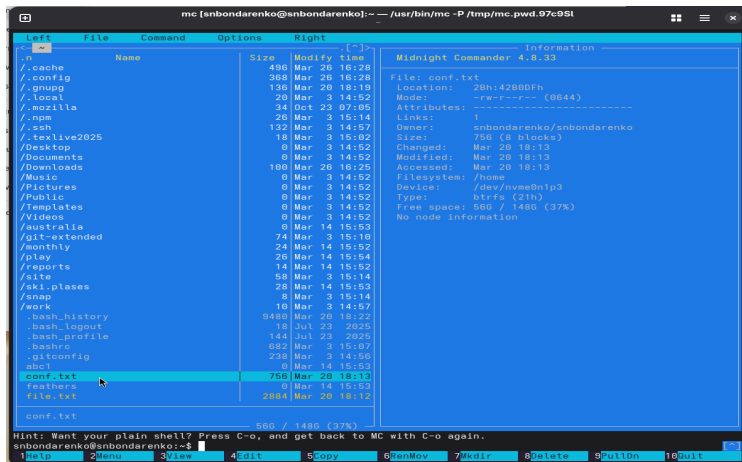


Рисунок 8: Информация

Работа с Midnight Commander

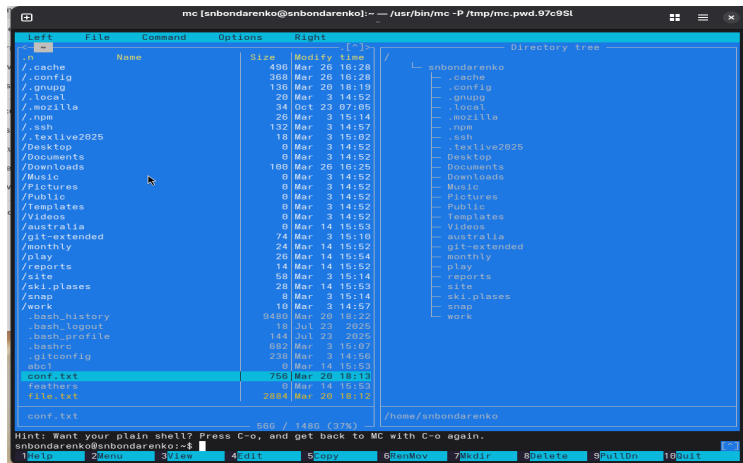
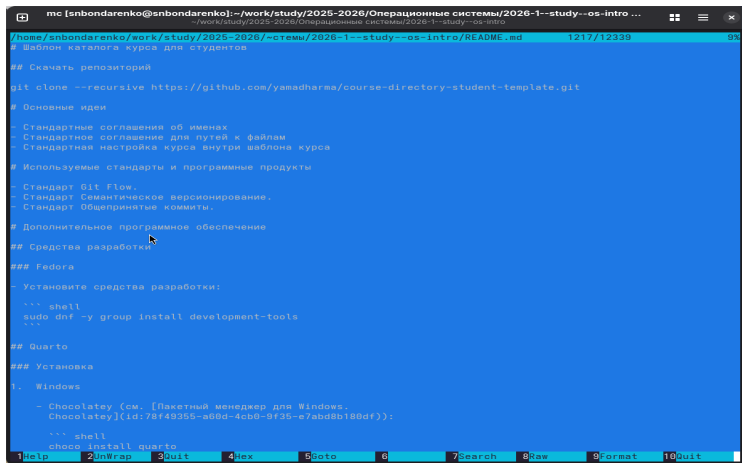


Рисунок 9: Дерево каталогов

Работа с Midnight Commander



The screenshot shows the Midnight Commander (MC) interface. The title bar indicates the current directory is `~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro ...`. The main window displays the contents of the file `README.md` in a blue-themed text editor. The file content includes instructions for cloning a repository, setting up development tools on Fedora and Windows, and installing Quarto. The bottom status bar shows various keyboard shortcuts for navigation and editing.

```
mc [snbondarenko@snbondarenko]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro ...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
/home/snbondarenko/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/README.md 1217/12339 9%
# Шаблон каталога курса для студентов

## Скачать репозиторий

git clone --recursive https://github.com/yamadharma/course-directory-student-template.git

# Основные идеи

- Стандартные соглашения об именах
- Стандартное соглашение для путей к файлам
- Стандартная настройка курса внутри шаблона курса

# Используемые стандарты и программные продукты

- Стандарт Git Flow.
- Стандарт Семантическое версионирование.
- Стандарт Общепринятые коммиты.

# Дополнительное программное обеспечение

## Средства разработки

### Fedora

- Установите средства разработки:

''' shell
sudo dnf -y group install development-tools
'''

## Quarto

### Установка

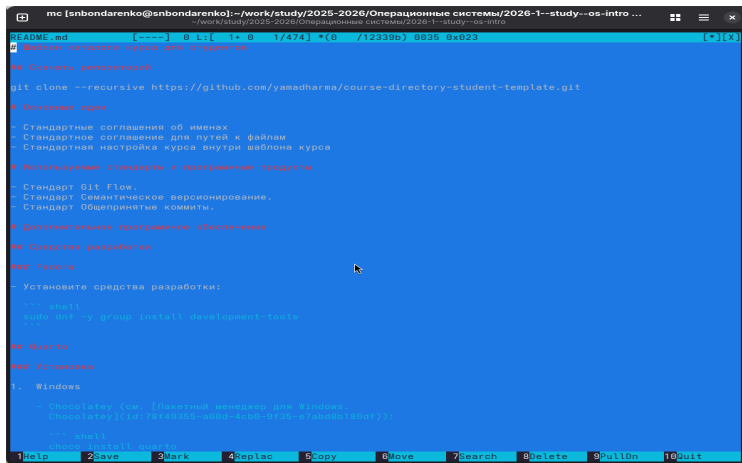
1. Windows

- Chocolatey (см. [Пакетный менеджер для Windows.
Chocolatey](id:78f49355-a60d-4cb0-9f35-e7abd8b180df)):

''' shell
choco install quarto
```

Рисунок 10: Просмотр содержимого текстового файла

Работа с Midnight Commander



The screenshot shows the Midnight Commander (MC) interface. The title bar indicates the current directory is `~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro ...`. The main window displays the contents of a `README.md` file. The file content includes instructions for cloning a repository, setting up standards, and installing development tools. The interface features a blue background with white text for the file content and red text for section headers. A status bar at the bottom shows various keyboard shortcuts like `1Help`, `2Save`, `3Mark`, etc.

```
mc [snbondarenko@snbondarenko]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro ...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
README.md [-----] 0 L: [ 1+ 0 1/474 ] *(0 /12339b) 0035 0x023 [*][X]
# Базовый шаблон курса для студентов

## Стандарты, рекомендуемый

git clone --recursive https://github.com/yamadharma/course-directory-student-template.git

## Основные идеи

- Стандартные соглашения об именах
- Стандартное соглашение для путей к файлам
- Стандартная настройка курса внутри шаблона курса

## Используемые стандарты и программные продукты

- Стандарт Git Flow.
- Стандарт Семантическое версионирование.
- Стандарт Общепринятые коммиты.

## Дополнительное программное обеспечение

## Средства разработки

### Fedora

- Установите средства разработки:

  ... shell
  sudo dnf -y group install development-tools

## Ubuntu

## Windows

1. Windows

- Chocolatey (cm, [Пакетный менеджер для Windows.
  Chocolatey] [id: 7d745355-460d-4c0b-9f35-e7add8b188df]):

  ... shell
  choco install powershell

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 11: Отредактируем содержимое текстового файла без сохранения результатов

Работа с Midnight Commander

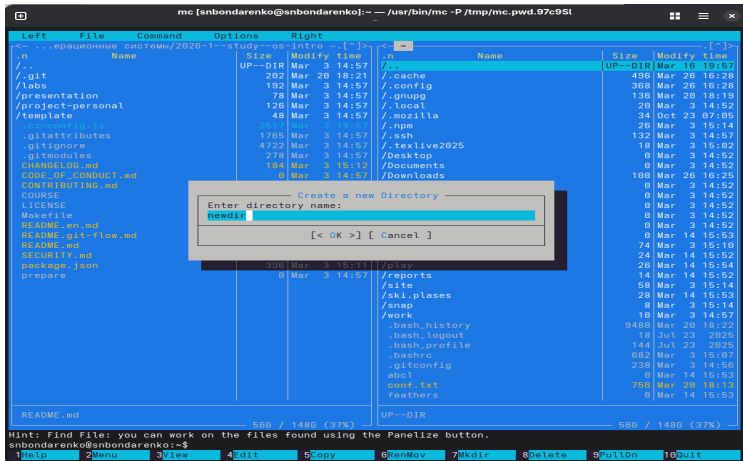


Рисунок 12: Создание каталога

Работа с Midnight Commander

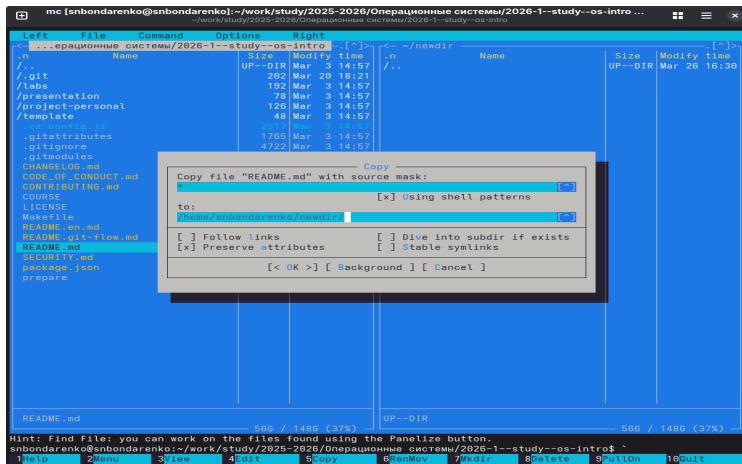


Рисунок 13: Копирование в файлов в созданный каталог

Работа с Midnight Commander

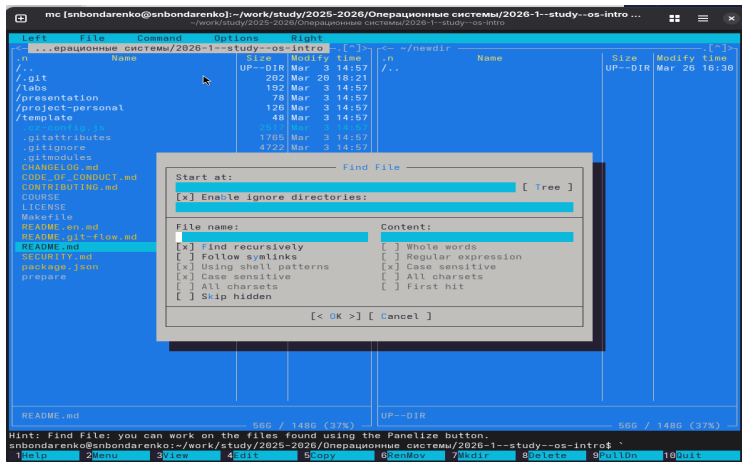


Рисунок 14: Поиск файлов

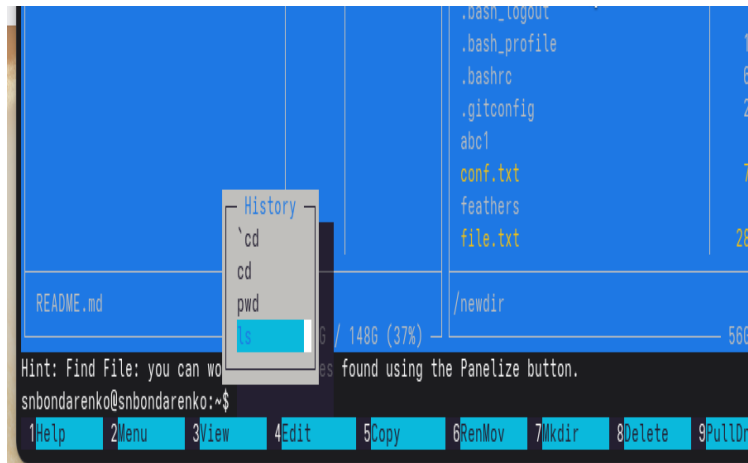


Рисунок 15: История команд

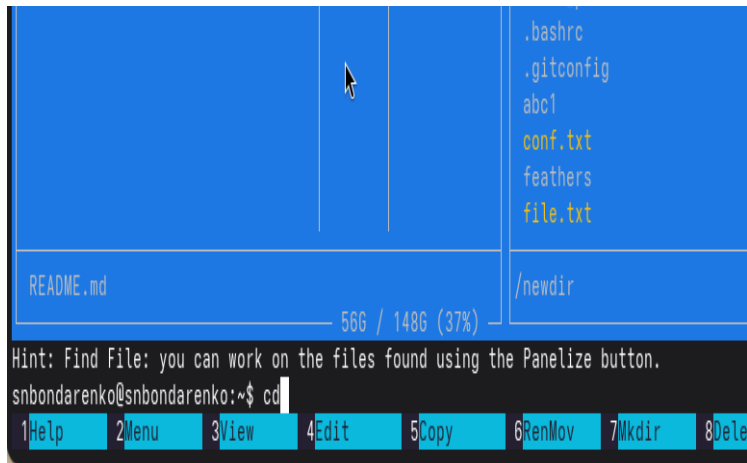
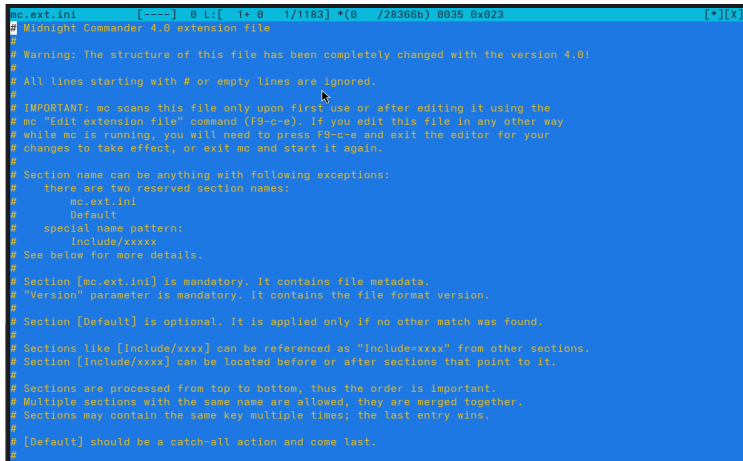
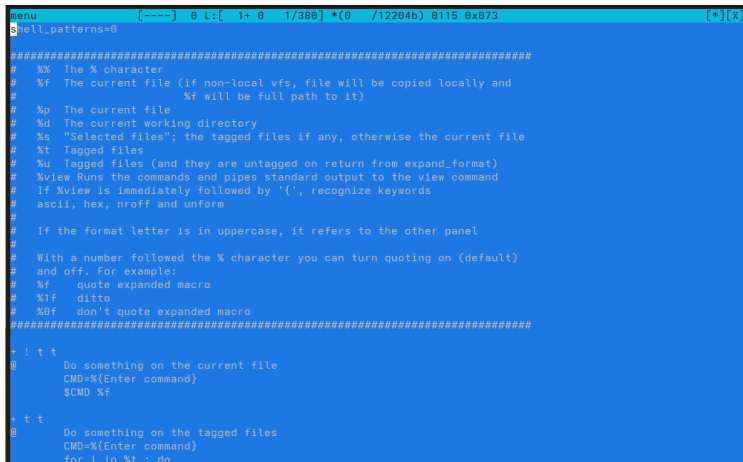


Рисунок 16: Переход в домашний каталог



```
mc.ext.ini      [----]  0 L:[ 1+ 0 1/1183] *(0 /28366b) 0035 0x023      [*][X]
# Midnight Commander 4.0 extension file
#
# Warning: The structure of this file has been completely changed with the version 4.0!
#
# All lines starting with # or empty lines are ignored.
#
# IMPORTANT: mc scans this file only upon first use or after editing it using the
# mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
# while mc is running, you will need to press F9-c-e and exit the editor for your
# changes to take effect, or exit mc and start it again.
#
# Section name can be anything with following exceptions:
#   there are two reserved section names:
#       mc.ext.ini
#       Default
#   special name pattern:
#       Include/xxxx
# See below for more details.
#
# Section [mc.ext.ini] is mandatory. It contains file metadata.
# "Version" parameter is mandatory. It contains the file format version.
#
# Section [Default] is optional. It is applied only if no other match was found.
#
# Sections like [Include/xxxx] can be referenced as "Include=xxxx" from other sections.
# Section [include/xxxx] can be located before or after sections that point to it.
#
# Sections are processed from top to bottom, thus the order is important.
# Multiple sections with the same name are allowed, they are merged together.
# Sections may contain the same key multiple times; the last entry wins.
#
# [Default] should be a catch-all action and come last.
```

Рисунок 17: Просмотр файла расширений

A screenshot of the Midnight Commander file menu. The window title bar shows 'menu' and '[---] 0 L:[1+ 0 1/380] *(0 /12204b) 0115 0x073 [X]'. The menu content is as follows:

```
shell_patterns=0

#####
# %% The % character
# %f The current file (if non-local vfs, file will be copied locally and
#      %f will be full path to it)
# %p The current file
# %d The current working directory
# %s "Selected files"; the tagged files if any, otherwise the current file
# %t Tagged files
# %u Tagged files (and they are untagged on return from expand_format)
# %view Runs the commands and pipes standard output to the view command
# If %view is immediately followed by '{', recognize keywords
#      ascii, hex, proff and unform
#
# If the format letter is in uppercase, it refers to the other panel
#
# With a number followed the % character you can turn quoting on (default)
# and off. For example:
# %f   quote expanded macro
# %if  ditto
# %0f  don't quote expanded macro
#####

+ ! t t
0   Do something on the current file
    CMD=%{Enter command}
    $CMD %f

+ t t
0   Do something on the tagged files
    CMD=%{Enter command}
    for i in %t ; do
```

Рисунок 18: Просмотр файла меню

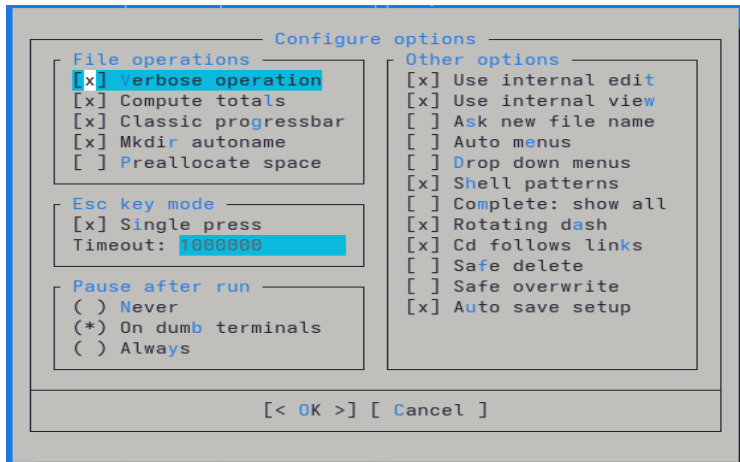


Рисунок 19: Конфигурация

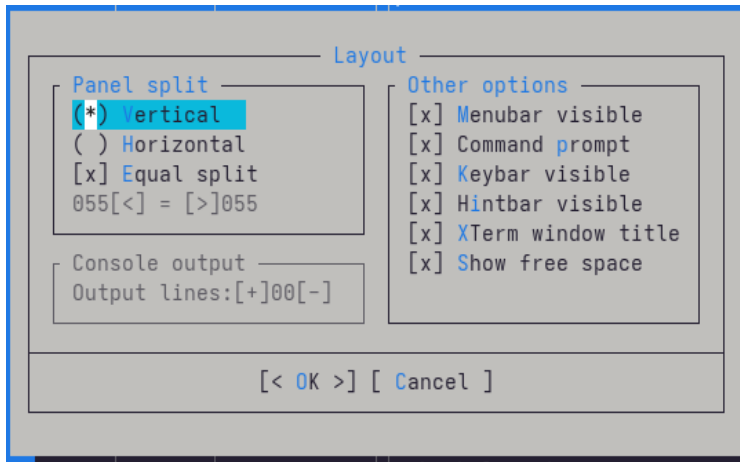


Рисунок 20: Внешний вид

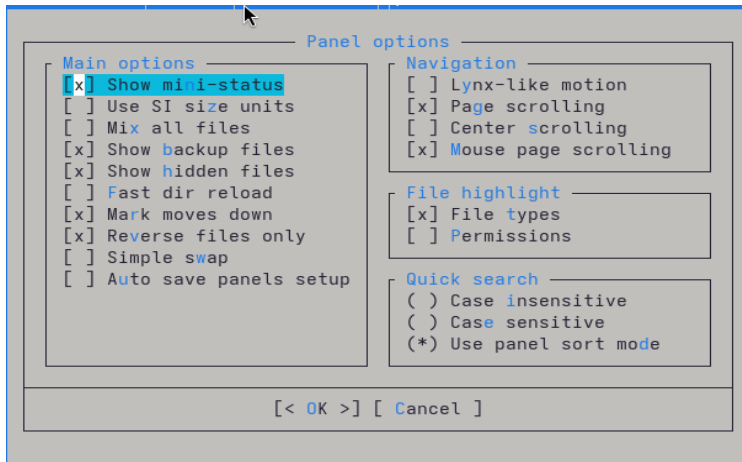


Рисунок 21: Настройки панелей

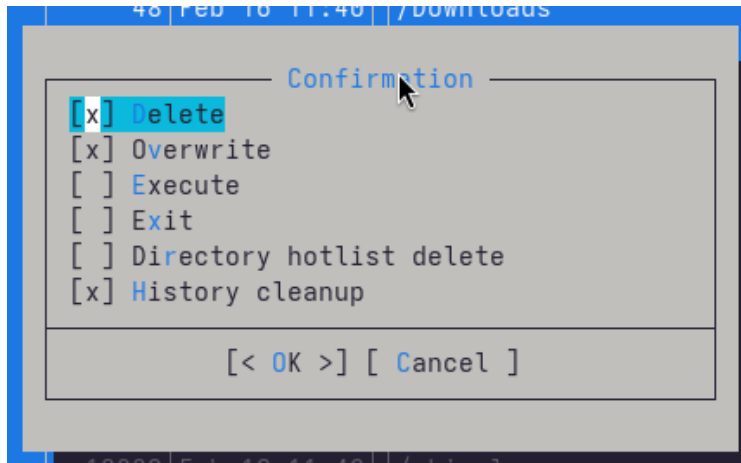


Рисунок 22: Подтверждение

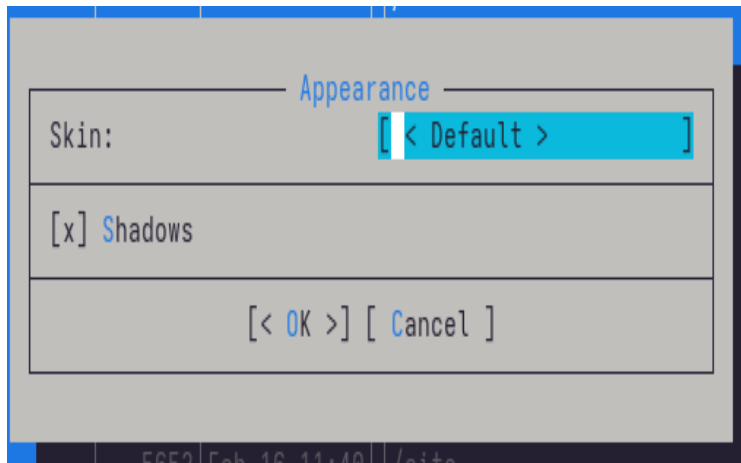


Рисунок 23: Оформление

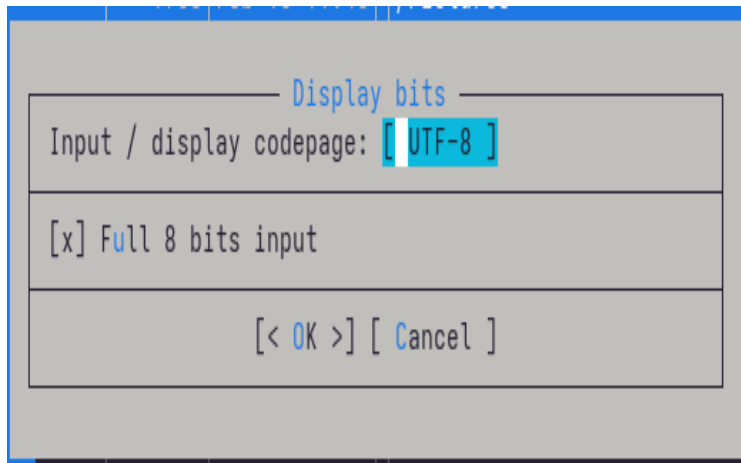


Рисунок 24: Кодировка символов

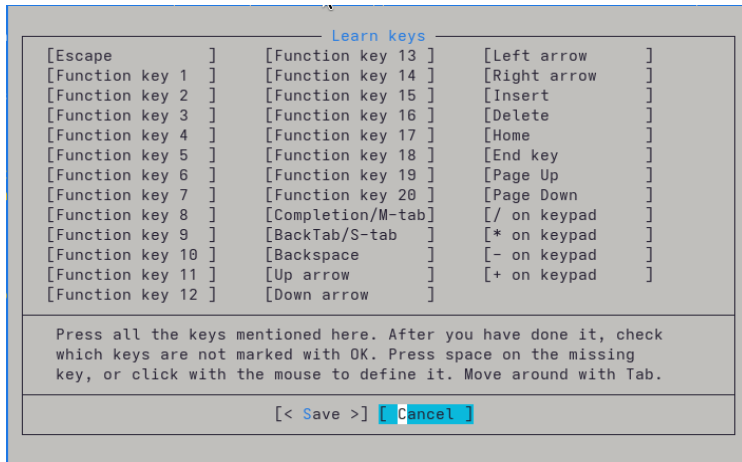
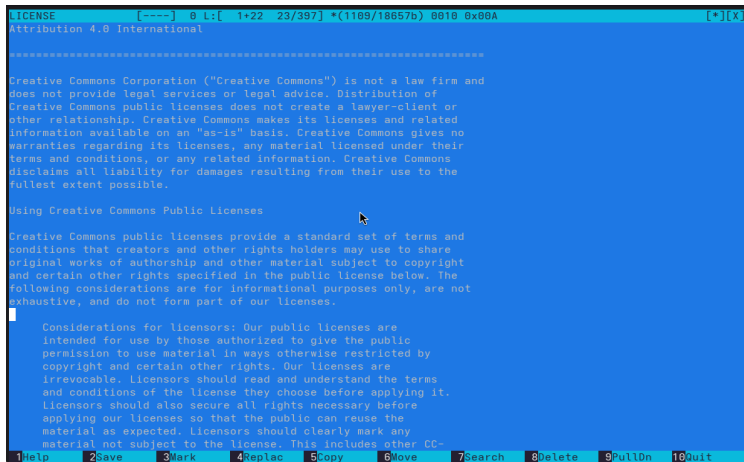


Рисунок 25: Распознавание клавиш



```
LICENSE      [-----]  8 L:[ 1+22 23/397] *(1109/18657b) 0010 0x00A      [*][X]
Attribution 4.0 International

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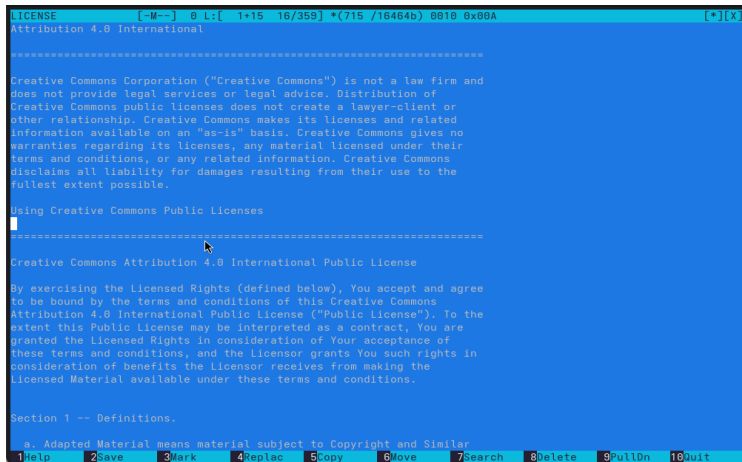
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  Licensors should also secure all rights necessary before
  applying our licenses so that the public can reuse the
  material as expected. Licensors should clearly mark any
  material not subject to the license. This includes other CC-
```

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit

Рисунок 26: Файл с текстом



```
LICENSE      [-M--]  8 L:[ 1+15 16/359] *(715 /16464b) 0010 0x00A      [*][x]
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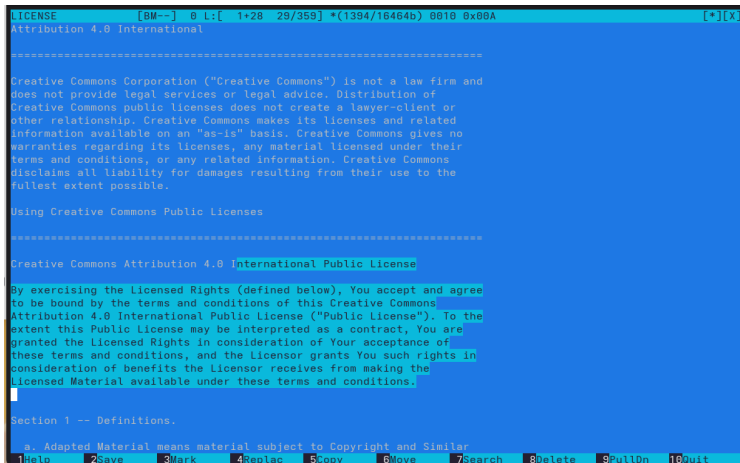
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a. Adapted Material means material subject to Copyright and Similar

1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 27: Файл с текстом



```
LICENSE      [BM--]  0 L:[ 1+28 29/359] *(1394/16464b) 0010 0x00A      [*][X]
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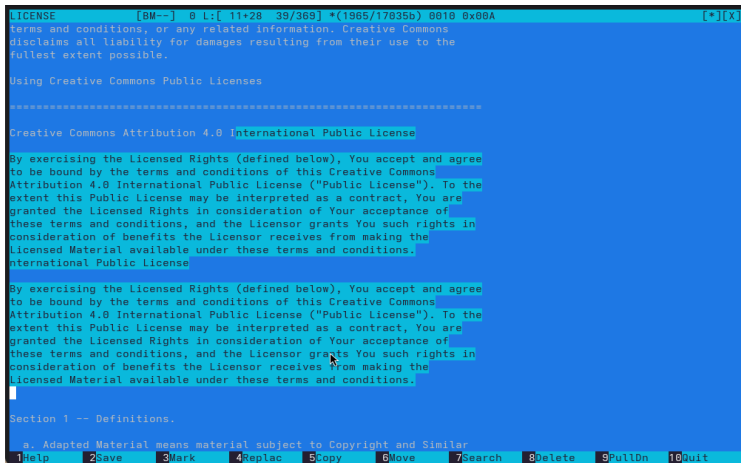
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  a. Adapted Material means material subject to Copyright and Similar

1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullOn     10Quit
```

Рисунок 28: Копирование фрагмента



```
LICENSE      [BM--]  0 L:[ 11+28  39/369] *(1965/17035b) 0010 0x00A      [*][X]
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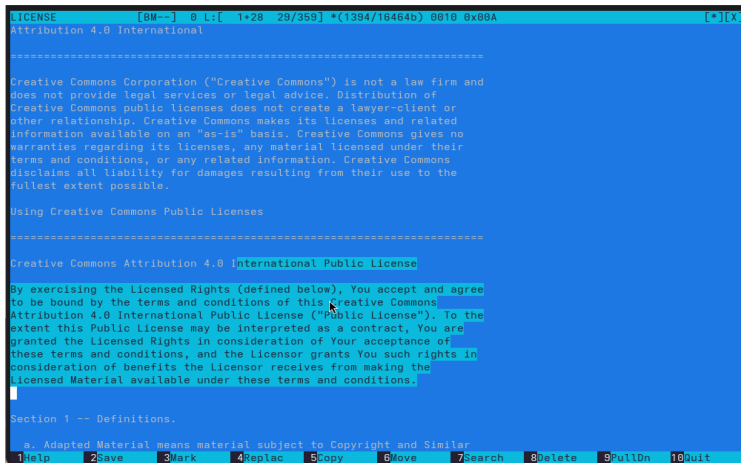
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1Help  2Save  3Mark  4Replac  5Copy  6Move  7Search  8Delete  9PullDn 10Quit
```

Рисунок 29: Сохранение

Работа с редактором Midnight Commander



```
LICENSE      [BM--]  0 L:[ 1+20  29/359] *(1394/16464b) 0010 0x00A      [*][X]
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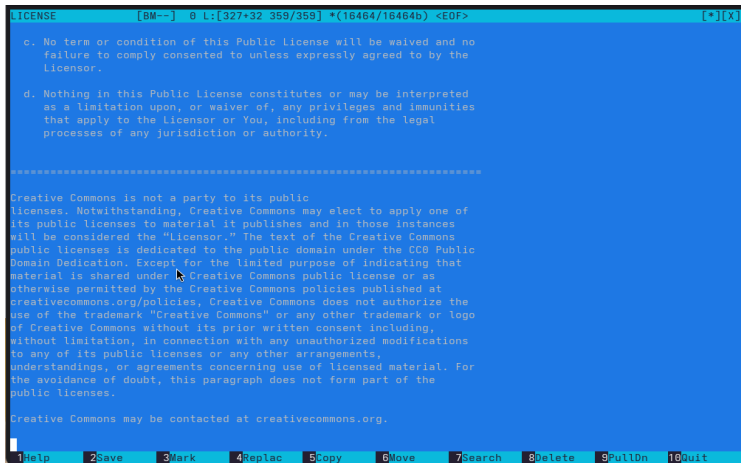
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1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 30: Отмена



```
LICENSE      [BM--]  0 L:[327+32 359/359] *(16464/16464b) <EOF>      [*][X]

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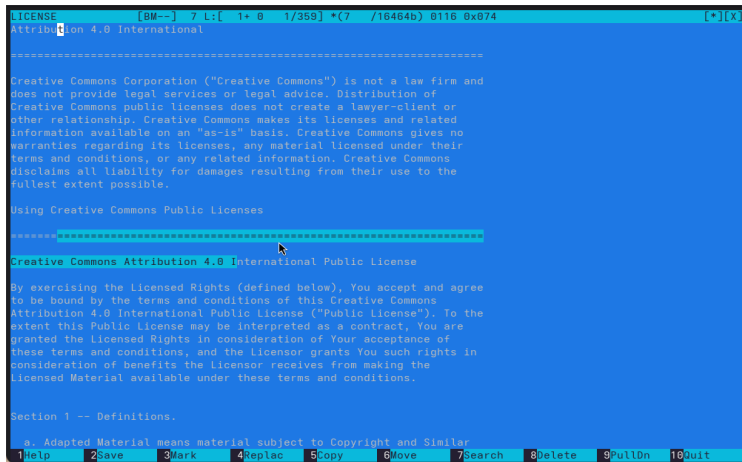
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1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 31: Переход в конец файла



```
LICENSE      [BM--] 7 L:[ 1+ 0 1/359] *(7 /16464b) 0116 0x074      [*][x]
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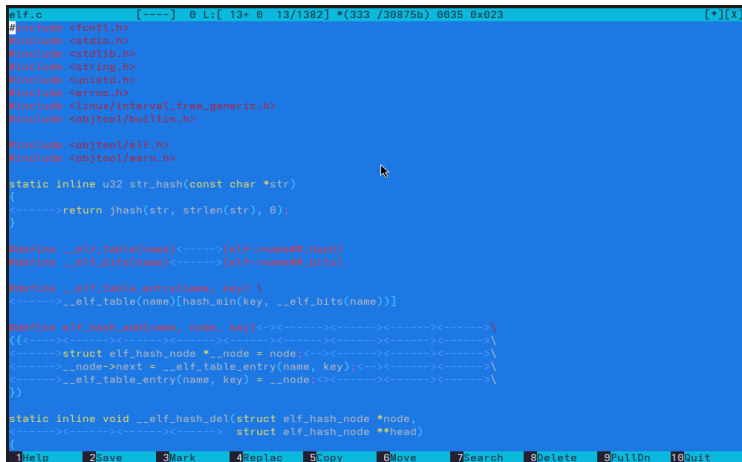
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Section 1 -- Definitions.

a. Adapted Material means material subject to Copyright and Similar

1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn      10Quit
```

Рисунок 32: Переход в начало файла



```
elf.c [----] 0 L: [ 13+ 0 13/1382] *(333 /30875b) 0035 0x023 [*][X]
#include <fcntl.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <errno.h>
#include <linux/interval_tree_generic.h>
#include <objtool/builtin.h>

#include <objtool/elf.h>
#include <objtool/warn.h>

static inline u32 str_hash(const char *str)
{
    ----->return jhash(str, strlen(str), 0);
}

#define __elf_table(name)----->{elf->name##_hash}
#define __elf_bits(name)----->{elf->name##_bits}

#define __elf_table_entry(name, key) \
    ----->__elf_table(name)[hash_min(key, __elf_bits(name))]

#define elf_hash_add(name, node, key)----->----->----->----->\
    ({----->----->----->----->----->\
    ----->struct elf_hash_node *__node = node;----->----->----->----->\
    ----->__node->next = __elf_table_entry(name, key);----->----->----->----->\
    ----->__elf_table_entry(name, key) = __node;----->----->----->----->\
    })

static inline void __elf_hash_del(struct elf_hash_node *node,
    ----->----->----->-----> struct elf_hash_node **head)
```

Рисунок 33: Файл с программой

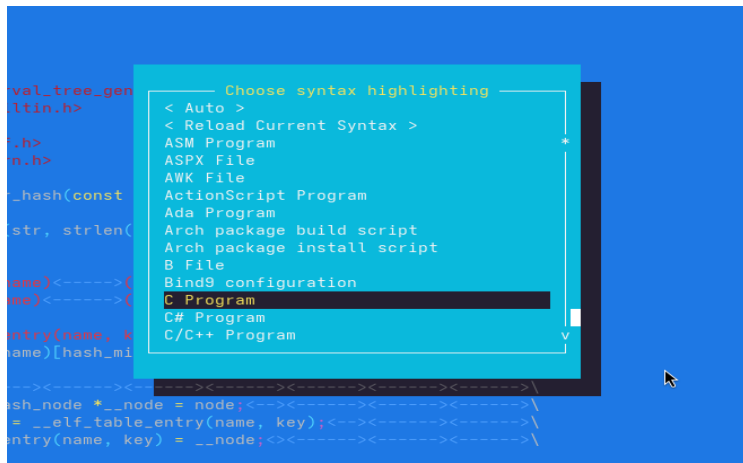


Рисунок 34: Цветовыделение синтаксиса

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.